Industry-Academia Tour

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Economics and Computation Lab,
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National Taiwan Ocean University

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Outline

1. Self Introduction
2. Teaching Courses
3. Focus of the Junior Project



https://josephcclin.github.io

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My research interests include algorithmic game theory, computational social choice, randomized algorithms, online learning and other theoretical computer science aspects.



Me @ University of Tokyo, January 2024.

Welcome to our lab:
Economics & Computation Laboratory

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Background | Research | CV | Teaching | Publications | Interest

Education

- BS.: Mathematics, National Cheng Kung University
- MS.: CSIE, National Chi Nan University
  - Supervisor: R. C. T. Lee
    Algorithms
- Ph.D.: CSIE (2011), National Chung Cheng University
  - Supervisors: Maw-Shang Chang & Peter Rossmanith
    FPT + Randomized Algorithms



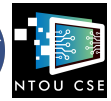
Work Experience

- Postdoc in Academia Sinica (2011–2018).
- *Quantitative Analyst* (intern) @ Point72/Cubist Systematic Strategies (2018–2020).
- *Quantitative Analyst* @ Seth Technologies Inc. (2020–2021).
- Assistant Professor @ Dept. CSIE, Tamkang University (2021–2024).
- Assistant Professor @ Dept. CSE, National Taiwan Ocean University (2024–Present).



Research Interests

- **Algorithmic Game Theory**
  - Equilibrium Computation, Computational Social Choice, Mechanism Design, etc.
- **Machine Learning Theory**
  - Online Learning with No-Regret, Bandit Problems, etc.
- **Design of TCS Algorithms**
  - Randomized Algorithms, Fixed-Parameter Algorithms, Approximation Algorithms, Online Algorithms.

Ongoing Projects

- *Fairness and Efficiency in Online Allocation of Indivisible Goods: Approximate Fairness, No-Regret Learning, and Optimization.*
  - Independent NSTC Project
  - Period: August 2026–July 2027.
- *Algorithmic Game Theory with Machine-Learned Predictions.*
  - Taiwan (NSTC) $\leftrightarrow$ Netherlands (NWO)
  - Period: July 2025–August 2026.



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Teaching Courses

- Undergraduates:
  - Introduction to Programming (II) [EMI]
  - **Data Structures**
- Graduates:
  - **Economics and Computation** [EMI]
  - Randomized Algorithms [EMI]
  - Mathematics for Machine Learning [EMI]



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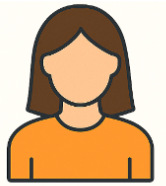
3 Focus of the Junior Project



Some fairness concepts







Alice

😍 😍 😍

😍 😍

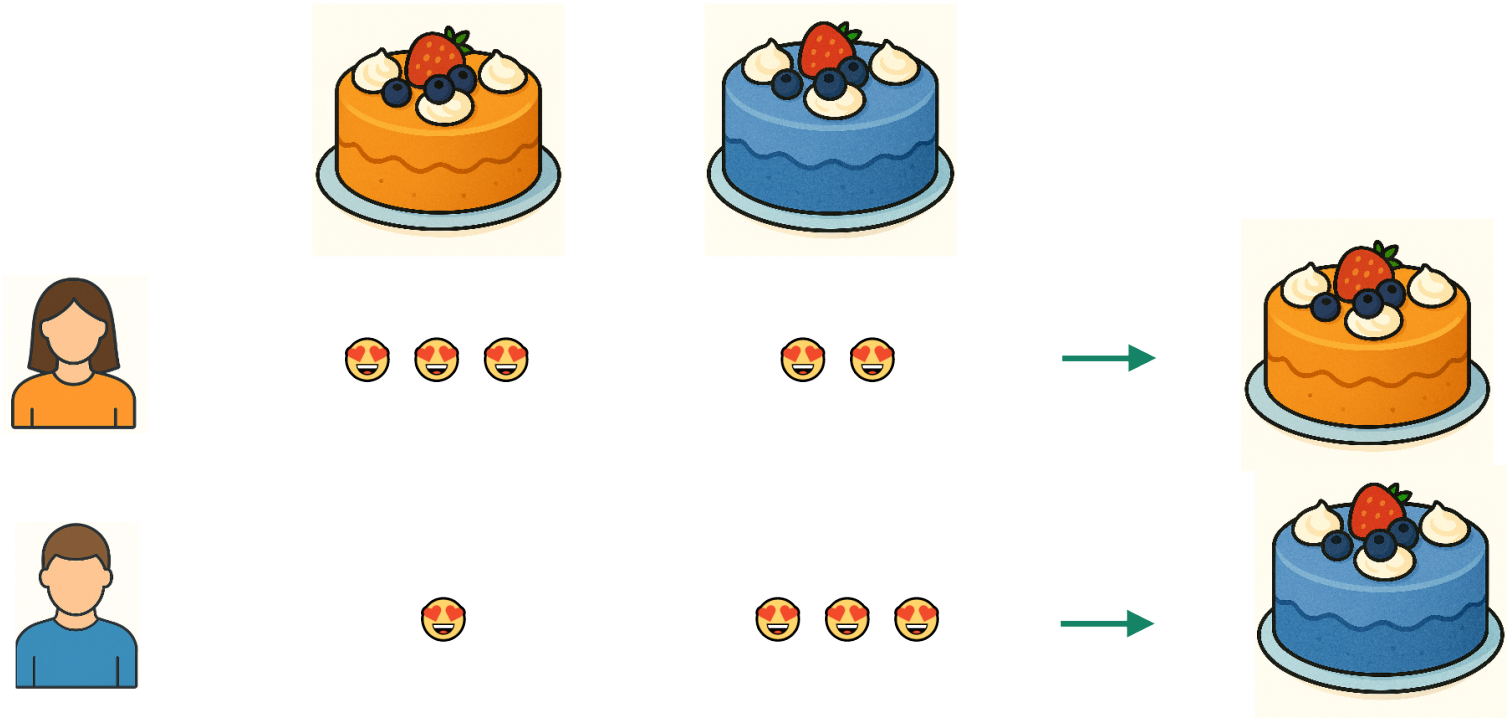


Bob

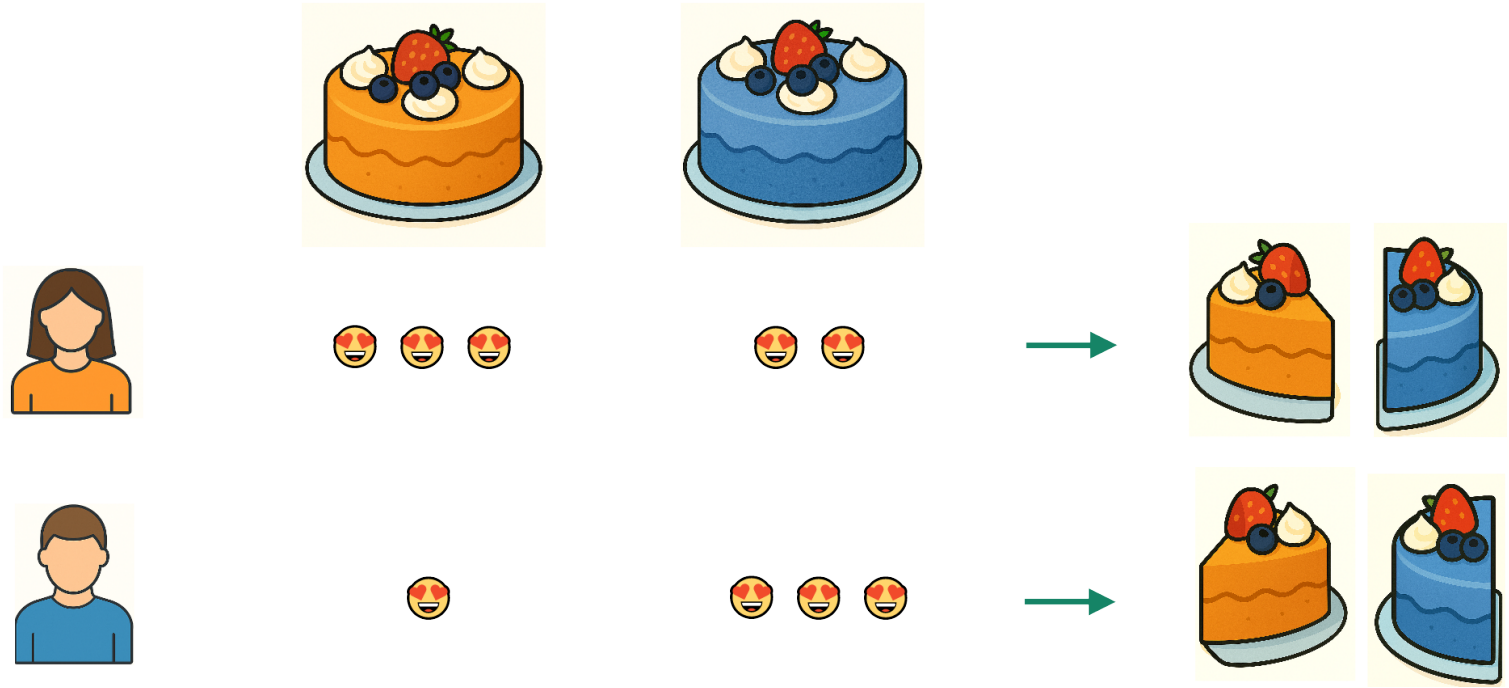
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Some fairness concepts



Some fairness concepts



Envy-freeness for allocating indivisible goods

Two-Partition Problem

Given a multiset S of positive integers, determine if it is possible to partition S into two disjoint subsets, say S_1 and S_2, such that the sum of the integers in S_1 is equal to the sum of the integers in S_2.

- $S = \{1, 5, 11, 5\}$

Envy-freeness for allocating indivisible goods

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Envy-freeness for allocating indivisible goods

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- $S = \{1, 5, 11, 5\}$
- $S_1 = \{11\}$,
  $S_2 = \{1, 5, 5\}$.

- $S = \{3, 5, 8, 10, 11, 14, 17, 19, 21, 22, 25, 33\}$.
- $S_1 = \{33, 25, 22, 14\}$,
  $S_2 = \{3, 5, 8, 10, 11, 17, 19, 21\}$.



Envy-freeness for allocating indivisible goods

NP-complete

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  $S_2 = \{3, 5, 8, 10, 11, 17, 19, 21\}$.



References (Lectures)

- Cake Cutting.
- Indivisible goods allocations.



Recent Progress

- CFD@IJCAI'26.
- TAAI'26 (Submitted).





Questions?

